Rowan University

# CS 03500 - Foundations of Cybersecurity

Cyberwarfare: Information Operations in a Connected World Projects

Project Part 4:

Examining Mission Assurance and Operational Procedures

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## 1. Mission Assurance Processes

Mission assurance for power grid computer networks is crucial for ensuring continuous operation, particularly in the face of cyberattacks, physical disruptions, and system failures. This process involves several layers of security and resilience strategies to protect the grid and its infrastructure. Here's a breakdown of the key components of mission assurance in this context:

Mission assurance for power grids centers on ensuring the availability, integrity, and confidentiality of network systems even in the event of an attack or disruption. Critical to this is the implementation of redundancy in power distribution, robust contingency planning, and incident response protocols. This involves maintaining backup power systems, like emergency generators, that ensure grid functionality during prolonged outages. The National Institute of Standards and Technology (NIST) offers guidelines on cybersecurity for industrial control systems (ICS) and emphasizes the need for mission assurance in protecting the power grid from disruptions, including backup power systems and disaster recovery plans (Stouffer et al.).

## 2. Industrial Control System (ICS)

Power grid networks often rely on ICS, which are used for monitoring and controlling grid operations, including the management of power generation and distribution. Securing these systems is crucial, as they are vulnerable to cyberattacks, which can compromise the grid's functionality. Implementing specialized defense technologies, such as network segmentation, firewalls, and intrusion detection systems, is vital to protect ICS components from cyber threats (Stouffer et al.).

### 3. Network Defense Technologies

- **Firewalls**

  A firewall is a security tool that acts as a barrier between a secure internal network and an external, less secure network like the internet. It controls the flow of data by enforcing predefined security policies to prevent unauthorized access and safeguard against threats. Firewalls are essential in protecting networks from malicious activities and unauthorized intrusions ("Cisco Secure Firewall: First Line of Defense").

- **Intrusion Prevention Systems (IPS)**

  An Intrusion Prevention System (IPS) is a network security tool, available as either hardware or software, that continuously monitors network traffic for potential threats. When malicious activity is detected, it actively prevents it by blocking, reporting, or dropping the harmful traffic. The IPS helps safeguard networks from attacks by taking immediate corrective actions (*What Is Intrusion Prevention System? | VMware Glossary*).

- **Security Information and Event Management (SIEM)**

  Security Information and Event Management (SIEM) is a security solution that enables organizations to identify and respond to potential threats and vulnerabilities. By monitoring and analyzing security events in real time, SIEM helps prevent disruptions to business operations before they can escalate into significant issues (Ibm).

## 4. Network Operational Procedures

Effective network operational procedures ensure that the power grid can continue functioning securely under adverse conditions. This includes regular security audits, incident response drills, and continuous monitoring of vulnerabilities. Procedures also cover patch management, access control policies, and the use of encrypted communication within the network to prevent data leaks or unauthorized system access.

In conclusion, mission assurance for power grid computer networks focuses on ensuring the grid's continued operation through resilience strategies like redundancy, contingency planning, and backup power systems. Securing Industrial Control Systems (ICS) is vital, requiring technologies like firewalls, Intrusion Prevention Systems (IPS), and Security Information and Event Management (SIEM) for threat detection and response. Effective network operational procedures, including security audits, patch management, and encrypted communication, are also essential to protect the grid from vulnerabilities and disruptions, ensuring both security and availability.

# Works Cited


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